Bluestock Mutual Fund Analytics Capstone Project

Final Project Report

Submitted by:

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Technology Stack:

- Python
- SQLite
- Jupyter Notebook
- Power BI
- Git and Github

Project Components:

- ETL Pipeline
- Exploratory Data Analysis
- Performance Analytics
- Risk Metrics
- Advanced Analytics
- Power BI Dashboard

Submission Version: v1.0

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Project Objectives

The primary Objectives of this project aew:

1.Develop an automated ETL pipeline for mutual fund data processing.

2. Clean, validate, and store data in a structured database.

3.Perform exploratory data analysis to identify trends and patterns.

4.Calculate mutual fund performance metrics.

5.Evaluate portfolio risk using advanced risk measures.

6. Analyze investor behavior and SIP continuity.

7.Build an interactive Power BI dashboard for business users.

8. Generate actionable insights and recommendations based on analytical findings.

Fund Master dataset

Contains scheme-level information such as scheme name , fund house ,category ,benchmark ,expense ratio, and launch date.

NAV History Dataset

Contains daily Net Asset Value (NAV)records used for return calculations, performance analysis, and risk metric computation.

Investor Transaction Dataset

Contains benchmark market index data used for alpha ,Beta, Tracking Error ,and comparative performance analysis.

ETL Design

The Project Follows a structured ETL(Extract ,Transform, Load) process to ensure data quality and consistency.

Extract

Data was collected from multiple source files, including Fund Master, Nav History,

Investor Transactions, Portfolios Holdings ,and Benchmark datasets.

Transform

The transformation stage involved:

- Data cleaning and validation
- Handling missing values
- Standardizing column names and formats
- Removing duplicate records
- Data type conversion
- Feature engineering for analytics

Load

The cleaned datasets were loaded into a SQLite database an d exported to processed CSV file for further analysis and dashboard development.

ETL workflow:

Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand fund characteristics, investor ,behavior and overall market trends.

Fund Analysis

- Distribution of mutual fund categories.
- Comparison of assets under management (AUM).
- Analysis of scheme -level performance.

Investor Analysis

- State-wise investment distribution.
- City -tier investment trends.
- Age -group Participation analysis..
- Gender -based investment patterns.

SIP Analysis

- SIP contribution trends over time.
- Investor participation levels.
- Transaction frequency patterns.
  The EDA stage helped identify important trends and prepare the data for advance analytics.

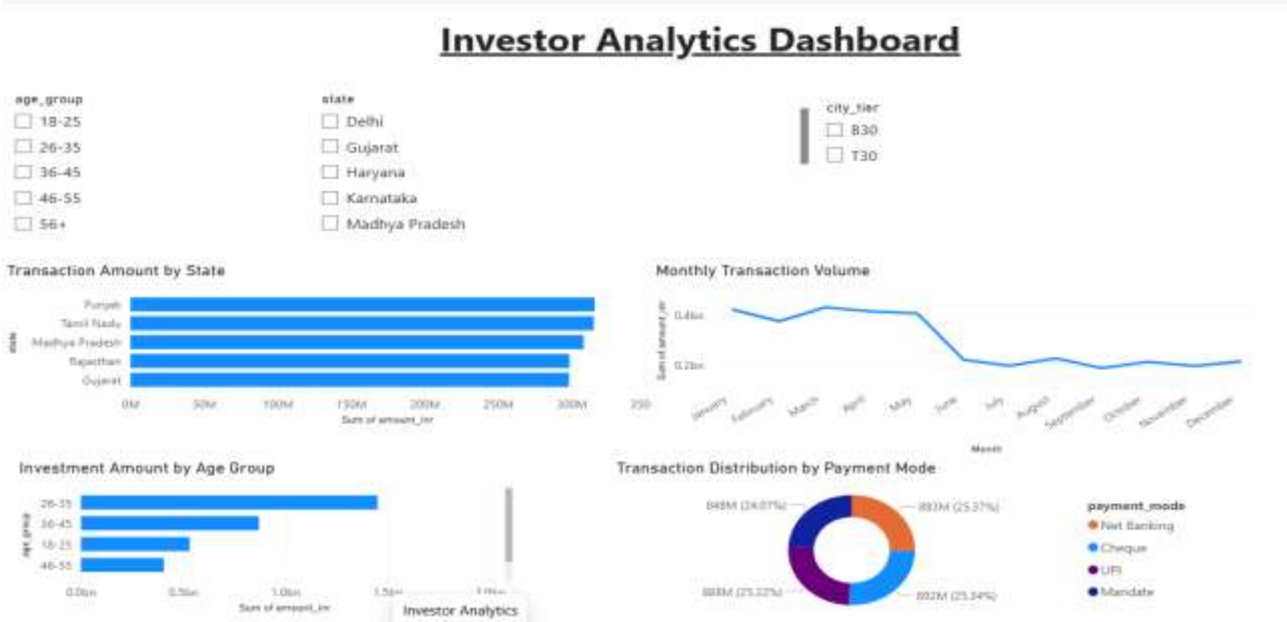


Fig2: Investor analytics dashboard

Key Observations:

1.Investors aged 26-35 contribute the highest investment amount.

2.Punjab and Tamil Nadu show the largest transaction volumes.

3.Transcation volume decreases after the first half of the year.

Advanced Analytics & Risk Metrics

Historical VaR & CvaR table.

amfi_code	VaR_95	CVaR_95
100016	-0.01436	-0.01806
100025	-0.00379	-0.00499
100033	-0.01903	-0.02346
101206	-0.01328	-0.01744
101207	-0.02602	-0.03246
101208	-0.00027	-0.00042
102885	-0.01261	-0.01549
102886	-0.01922	-0.02325
102887	-0.01523	-0.01941
118632	-0.01395	-0.01762
118633	-0.01344	-0.01663
118634	-0.02544	-0.0323
118635	-0.01255	-0.01618
118636	-0.0038	-0.00492
119092	-0.01375	-0.01733
119093	-0.01423	-0.01749

- VaR and CvaR quantify downside risk.
- Some schemes exhibit significantly larger potential losses than others.

Investor Cohort table.

```
cohort_analysis.head()
```

Python

	avg_sip_amount	total_invested
cohort_year		
2024	107422.541832	3491125187
2025	109158.577061	30455243

Observation:

- The 2024 cohort contributes the highest investment volume.

- Average SIP amount remains strong across cohorts.

SIP Continuity Analysis table.

```python
continuity.head()
```

investor_id	sip_count	avg_gap	status
INV000004	6	85.400000	At Risk
INV000008	6	70.400000	At Risk
INV000010	6	64.800000	At Risk
INV000011	7	40.166667	At Risk
INV000012	8	57.000000	At Risk

Observations:

- Several investors were flagged as “At Risk” due to gaps exceeding 35 days.
- SIP continuity monitoring can improve retention.

HHI concentration Analysis table

```python
hhi.head()
```

	amfi_code	HHI
0	100016	0.139534
1	100033	0.147592
2	101206	0.129332
3	101207	0.200700
4	102885	0.174709

- Higher HHI values indicate more concentrated portfolios.
- Concentration risk varies across schemes.

Fund Recommender Engine

```
Top Recommended Funds:

                                    scheme_name risk_grade  sharpe_ratio  \
14  ICICI Pru Liquid Fund - Regular - Growth         Low          7.68
23      Kotak Liquid Fund - Regular - Growth         Low          6.18
30       ABSL Liquid Fund - Regular - Growth         Low          5.14

    return_3yr_pct
14            7.68
23            6.18
30            5.14
```

- Low -risk funds with the highest Sharpe ratios are recommended.

Dashboard Overview

To support data -driven decision making, four interactive Power BI dashboards were developed as part of this project. These dashboards provide a comprehensive view of mutual fund industry trends, fund performance , investor behavior , and SIP activity.

The dashboard allows users to filter and analyze data across multiple dimensions, including fund house ,category ,risk level, investor demographics, geographic location , and market trends.

The dashboards developed are:

1.Industry Overview Dashboard

2.Fund Performance Dashboard

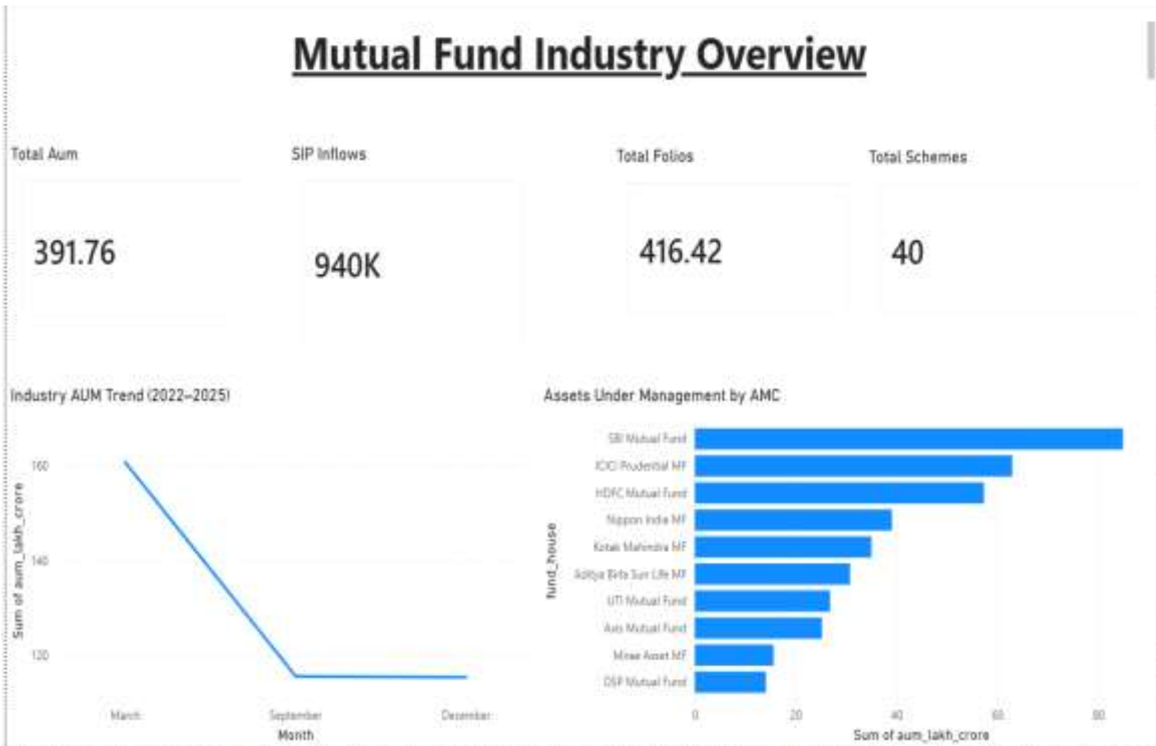
3. Investor Analytics Dashboard

4.SIP & Market Trends Dashboard

These dashboards transform complex analytical results into intuitive visualizations for easier interpretation and business decision-making.

Industry Overview Dashboard

The Industry Overview Dashboard provides a high -level view of the mutual fund industry and category-level trends.

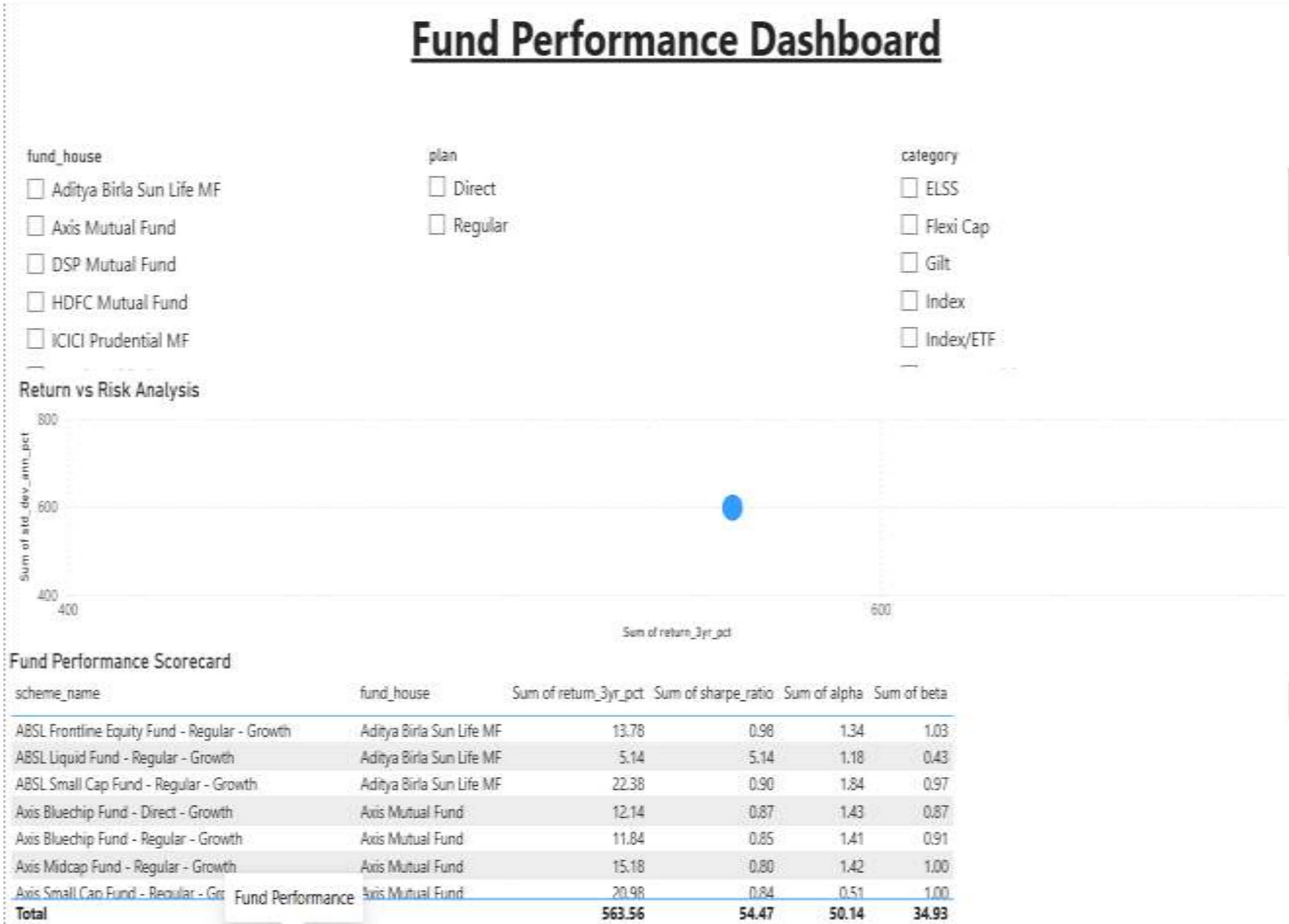


Key Observations

- Mutual fund assets are concentrated among a limited number of fund categories.
- Certain categories consistently attract higher investor participation and inflows.
- Industry -level analysis helps identify growth opportunities and changing investor preferences.
- Category comparisons reveal differences in fund popularity and market positioning

Fund Performance Dashboard

The fund Performance Dashboard evaluates mutual fund performance using multiple return and risk-adjusted metrics.



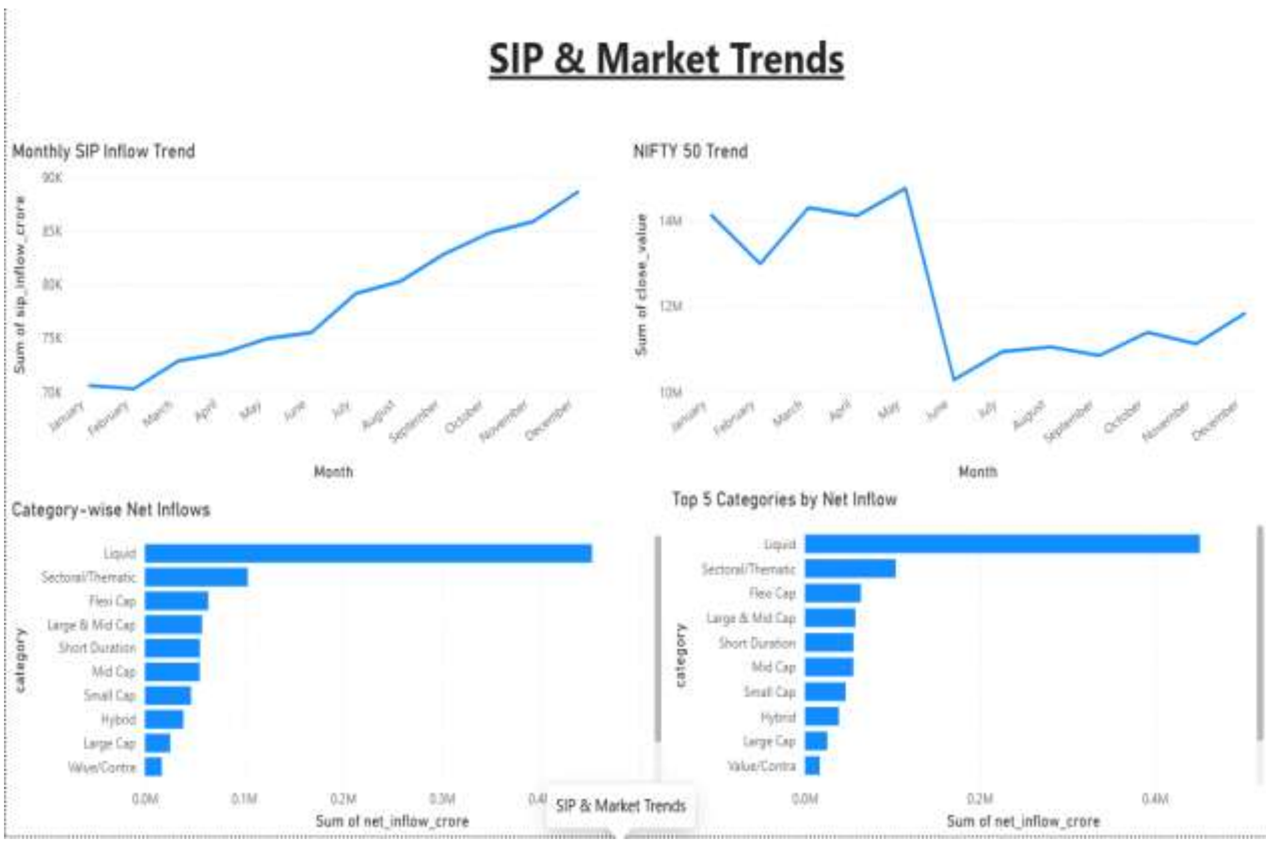
Key Observations

- Fund returns vary significantly across schemes and categories.
- Higher -return funds generally exhibit higher volatility and risk.

- Sharpe Ratio analysis helps identify funds with superior risk-adjusted performance.
- Alpha and Beta metrics provides insights into market sensitivity and benchmark outperformance.
- Scatter plot analysis demonstrates the relationship between return and risk.

SIP & Market Trends Dashboard

The SIP & market trends dashboard examines systematic investment behavior and market-related fund flows.

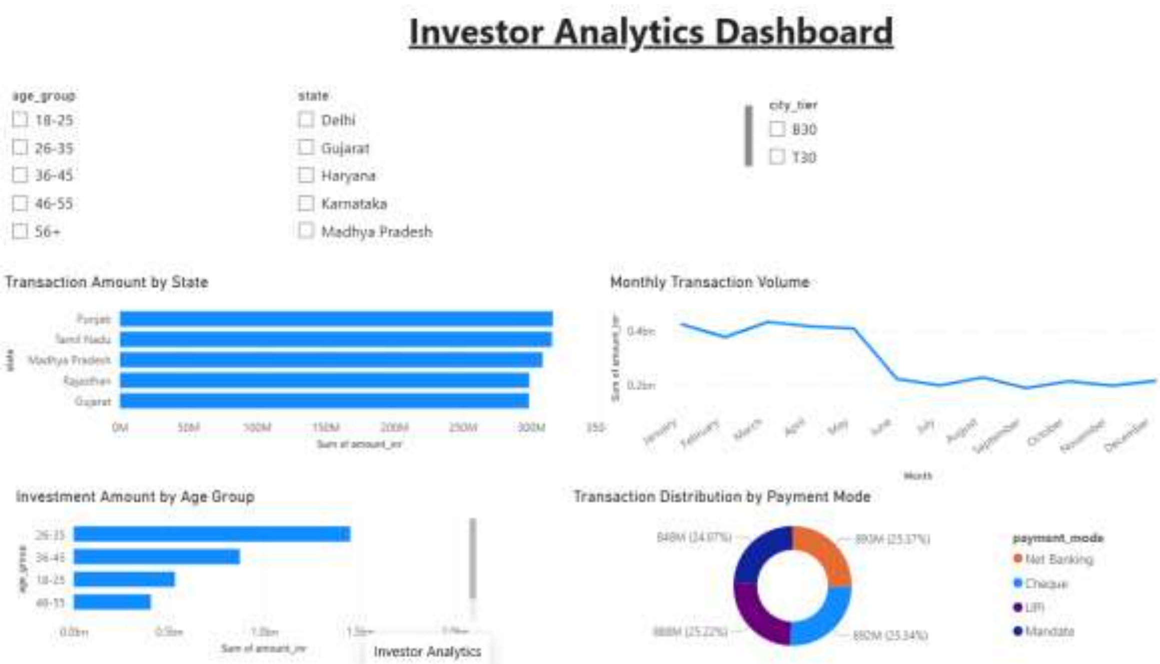


Key Observations

- Monthly SIP inflows demonstrate an overall increasing trend.
- Liquid funds receive the highest category-wise net inflows.
- Market conditions influence investor participation and contribution patterns.
- Category -level inflow analysis highlights investor preferences across fund types.
- Long -term SIP growth indicates increasing investor confidence

Investor Analytical Dashboard

The investor analytical dashboard analyzes investor demographics ,geographics distribution, and transaction behavior.



Key Observations

- Investors aged 26-35 contribute the highest transaction amounts among the analyzed states.
- Monthly transaction activity declines during te second half of the year.
- Multiple payment modes are actively used , with relatively balanced transaction distribution.
- Geographic and demographic segmentation provides valuable insights into investor behavior.

Key Findings

- Investors aged 26-35 represent the largest contributor to total investment volume.
- Punjab and Tamil Nadu emerged as the leading states in transaction activity.
- The 2024 investor cohort contributes the highest overall investment amount.
- Several investors were identifies as “At Risk “due to irregular SIP contribution intervals.
- Historical VaR and CvaR analysis shows variation in diversification funds.
- Low -risk liquid funds achieved the highest Sharpe ratio within low-risk category.
- Dashboard analysis confirmed strong growth in SIP participation and category level inflows.

Limitations and Recommendations

Limitations

- The analysis is based on historical data and may not reflect future market conditions.
- Certain datasets contain limited historical coverage.
- Real-time market feeds were not integrated into the system.
- Investor transaction behavior was analyzed using available records only.
- External economic and market factors were not explicitly modeled.

Recommendations

- Integrate live NAV and market data feeds for real -time analytics.
- Develop machine learning-based recommendations models.
- Enhance investor segmentation using behavioral analytics.
- Implement automated alerts for SIP continuity monitoring.
- Expand risk analysis using additional portfolios and market indicators.
- Publish dashboards through Power BI for wider accessibility.

Conclusion

The Bluestock Mutual Fund Analytics Capstone project successfully demonstrates the application of data engineering , analytics , risk assessment , and business intelligence techniques in the mutual fund domain.

The project involved the complete analytics lifecycle, including data ingestion , cleaning , validation, database creation, exploratory data analysis, performance evaluation , advanced risk analytics , and dashboard development . Multiple Error, and Maximum Drawdown were calculated to evaluate mutual fund performance. Advanced analytics techniques including Value at Risk(VaR), Conditional Value at Risk(CvaR) , investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration Analysis(HHI), and a fund recommendation System provided deeper insights into investment behavior and risk exposure.

Interactive Power BI dashboards were developed to visualize industry trends, fund performance, investor demographics, and SIP activity.

Overall, the project highlights the importance of data -driven decision making in the financial services industry and provides a scalable framework for future mutual funs analytics and investment research.